Team: Background Checkers
Team members: Anthony Man, Amari Thomas, Braeden Alonge, Max Schemenauer
Assignment: Project Overview, Background, Difficulty and Relevance
Course: CSC 5870 - LLM Post-Training & Retrieval
Date: 09/10/2026

**Project overview and user need**

On the first day of class, we discussed the trustworthiness of LLMs given various scenarios. In all cases, there were substantial doubts as to whether or not the given scenario (such as if the LLM provided links, stated a familiar fact confidently, etc.) was sufficient for us to trust the LLM output. This project provides a solution for this problem, using an LLM as a quick background check for LLM output (this is extendable to other source text). The basic idea for this project is that we use a lightweight LLM to summarize a piece of text (our main target is another LLM's output), grabbing the main concepts.

The LLM will then utilize an internet search tool to fetch reputable articles/pages related to the topic. After it has obtained several pages, it will summarize the contents of those pages and compare that with the content of the target text. If there are notable differences in sentiment or information, the LLM will flag the target text as being different from current online discourse on that topic, and provide the sources it pulled. This system, ideally speaking, would be implemented as a web extension and would passively scan LLM (or other desired text), checking for consistency with relevant online sources.

This design facilitates the fact-checking of LLM output by providing a cursory preliminary screening of all text so that the user does not need to check every assertion on their own. It also avoids the problem of trustworthiness that we previously discussed because it does not provide any assertions of fact, but rather leaves it up to the user by providing further reading on the topic.

**Background related work and baseline**

The proposed project relates to research on LLM hallucination detection, retrieval-augmented generation (RAG), automated fact-checking, and natural-language inference. LLMs produce confident and well-written responses even if the information given is incorrect, incomplete, or outdated. The goal of our project is to address this problem by using a

system to scrape online sources and perform a preliminary consistency check. Retrieval-augmented generation will be especially important to this project. RAG systems work using search tools to query external data, such as web pages and databases, retrieving relevant information that undergoes pre-processing, including tokenization. Once the pre-processed information is retrieved, it is supplied to the pre-trained LLM, allowing for it to generate more grounded answers.[1] For our project, retrieval will be used differently, instead of retrieving sources before an answer is written, the system will retrieve sources after target text has already been created in order to evaluate whether that text is consistent with current online information.

SelfCheckGPT is a related hallucination-detection method that generates several responses to the same prompt, measuring whether those responses agree with one another. This will serve as a useful baseline because it can identify statements that a model may be uncertain about without requiring internet access.[2] Another related research method is Chain-of-Verification (CoVe). CoVe works by taking an LLM initial response and creating questions to verify the important claims in that response. It answers those questions independently and uses the results to revise the original response. [3] Both of these research methods will serve as a baseline for our project. Our project differs by checking already-generated text against external online sources rather than relying only on additional LLM-generated responses. Instead of revising the target text, our system will flag possible inconsistencies and provide the user with relevant sources for further review.

A credible simple baseline for this project would be extracting the main topic from a target text, perform a single web search on it, and compare the target text to the first relevant source. This baseline would be a simple version of the project we want to complete. Our project will improve on this by evaluating multiple sources, assessing source quality, and clearly identifying which claims are not agreeing. The capability gap that this project will address is comparing target texts with multiple online sources. Other LLM hallucination checkers often compare the targeted LLM's text with another LLM or itself. Our project will scan reputable sources to give a more accurate way of checking for inconsistencies and hallucinations.

**Difficulty and feasibility**

The primary difficulty of this project lies in the very same inherent uncertainty in LLM's that we are aiming to reduce. Additionally, there are several difficulties relating to data access, evaluation, and a clear scope

In order to tune the LLM on how to properly parse web search API returns into something usable we need copious amounts of data. Due to the nature of our project this will likely need to be hand curated. Depending on what our web search API returns, we can likely find a dataset of similar content. However, efficiently parsing that content into a reliable background checking format may be needed. In theory, many LLM's can zero shot, or one shot (with a clean example), the extraction of relevant information from a web page, but this remains to be tested, and can be subject to unforeseen and untrackable errors. Additionally, even gaining access to a low-cost web search API, can be very difficult. As mentioned above, this project is susceptible to the very thing it is trying to help with. Our LLM's queried sources and written summaries are themselves not substantially different from what another LLM or a news article might output. The feasibility of this project relies on the fact that we can have more control and maintain a much tighter objective than a general LLM.

Another potential pitfall is the fact versus opinion debate. Deciding whether a claim is supposed to be an opinion or a fact in the first place is non-trivial. If we decide to check only facts or only opinions, we will have to reliably sort them. If we decide we don't care about the distinction, getting an accurate read on public sentiment becomes a biased gamble, especially if up-to-date news sources have shifted, but public sentiment simply hasn't caught up yet.Building on the previous risk, establishing a clear scope is critical. If our system is given a dense academic paper containing complex new knowledge, the LLM may not understand the claim well enough to search for something relevant. On the other hand, if it is given a niche, yet simple topic, it may not be able to find enough information to place the topic in context. We likely cannot background-check every single type of content without performance dropping significantly, so we should bound our scope to specific types of claims.

Lastly, evaluation is incredibly challenging. There are many parts of this system that could cause the end result to degrade in quality. This type of problem often requires subjective evaluation, or specialized NLP metrics for relevance or completeness. Certain components we can analyze include how well the LLM understood the initial claims, how relevant the retrieved sources are, and whether it misunderstood or improperly parsed the content from the sources.

**Relevance to CSC 5870**

This project is highly relevant to a lot of major topics in this class, especially retrieval, grounding, agentic workflows, and evaluation. The course is focused on combining LLMs with external tools and retrieved information to make systems that are more traceable, reliable, and

useful. Our project will follow that idea because we will use LLMs as a part of a larger verification pipeline instead of treating the model itself as the final factor for verification (leaving the user as the final source of truth).

Grounding and retrieval are a major part of the project because the system will search for relevant online sources and summarize found information, then compare that information with the original text. This is directly tied to the course's topics of retrieval-augmented generation, citation discipline and tool-based verification.

The project also focuses on agentic workflows because the system will have to complete several connected steps (like identifying important claims, generating searches, using external tools, selecting useful sources, and comparing retrieved information). Lastly, evaluation is also going to be important because we will need to measure how well the system performs on retrieving relevant sources and how accurately it identifies information that does not align with the selected text.

**Project plan and communication**

Phone numbers have been shared, and communication will take place within a group chat that includes all team members. After each class, members may also tag up and discuss required work to complete milestones on time. Responsibility will be determined during each correspondence, though it is likely that all members will simply work together on the action items laid out for the week. Project milestones are as follows (subject to change depending on weekly progress):

- Weeks 1-3: Group formation and project planning (already completed as of writing this document)
- Weeks 4-6: Preliminary research
    - Understand necessary techniques and tools
    - Research models and common approaches
    - Acquire access to needed tools and resources
- Weeks 7-8: Data collection
    - Formulate what data should look like
    - Obtain data as determined
- Weeks 9-10: Pipeline setup
    - Set up the pipeline, including model and necessary tools

    - Prepare model for training
- Weeks 11-12: Train and evaluate
    - Run training using data
    - Come up with evaluation metrics
    - Run inference on trained model and evaluate model performance
- Weeks 13-14: Improve model performance based
    - Based on evaluation, improve model
        - Tweak data
        - Update pivot on tools or model as needed
- Weeks 15:
    - Compile findings into poster presentation
    - Prepare demonstration

References

Note: AI was used to assist with grammar, spelling, and sentence clarity. Project ideas, research selection, and final content was developed and reviewed by the authors.

[1] What is Retrieval-Augmented Generation (RAG)?
https://cloud.google.com/use-cases/retrieval-augmented-generation

[2] SELFCHECKGPT:Zero-Resource Black-Box Hallucination Detection
for Generative Large Language Models
https://arxiv.org/pdf/2303.08896

[3] Chain-of-Verification Reduces Hallucinations in Large Language Models
https://arxiv.org/pdf/2309.11495